



4-Fluoro-1,3-dioxolan-2-one (FEC)

ELECTROLYTE ADDITIVE

PRODUCT OVERVIEW

4-Fluoro-1,3-dioxolan-2-one (FEC) is an electrolyte additive supplied for battery-electrolyte formulation, additive screening, and electrochemical research. This sheet summarizes specification limits and representative quality-control results.

PRODUCT INFORMATION

CAS No. 114435-02-8 | Formula $C_3H_3FO_3$ | Abbreviation FEC

TECHNICAL SPECIFICATIONS

PROPERTY	SPECIFICATION	REPRESENTATIVE QC RESULT	UNIT
Appearance	Colorless transparent liquid, no impurities	Colorless transparent liquid, no impurities	—
Assay	≥99.95	99.964	wt%
Water (Karl Fischer titration)	≤30	16.9	µg/g
Acidity, as HF	≤15	5.6	ppm
Color	≤10	6.4	Hazen
Sulfate	≤5	<5	ppm
Chloride	≤1	0.88	µg/g

REPRESENTATIVE QC HIGHLIGHTS

99.964 wt% assay

16.9 µg/g water

5.6 ppm acidity

RESEARCH APPLICATIONS

Electrolyte formulation development; additive screening; cell-material compatibility studies; and electrochemical research. Suitability should be confirmed in the intended system.

STORAGE / SHELF LIFE

180 days

QUALITY / DOCUMENTATION

Specification limits and representative quality-control results are provided for technical reference. Lot-specific Certificate of Analysis is available upon request.

Supplied and technically supported by Winigen Materials.

www.winigenmaterials.com | contact@winigenmaterials.com

FEC-TDS-001

Specification limits and representative quality-control results are provided for technical reference. Representative results are lot-specific and are not guaranteed values. The applicable lot-specific Certificate of Analysis governs material supplied.